

DEVICE SETUP RUNBOOK

Equipment Pre-Use Checklist System · Mobile fleet rollout

Belfast Coal Mine · Enaleni Engineering (Pty) Ltd

For whoever onboards the tablet fleet — IT supervisor, operations lead, or first-day rollout team.

1. Before you start

This runbook walks one person through enrolling the mine's mobile device fleet onto the Equipment Pre-Use Checklist System. Each tablet or phone takes about 10 minutes to provision the first time, including first sign-in and biometric enrolment. Twenty devices is a comfortable morning's work.

1.1 What you need on the desk

- **All target devices** — powered on, fully charged, with the OS-level biometric (fingerprint or face) already enrolled in device Settings. The Equipment Checklist app uses whatever the OS exposes, so this step is non-negotiable.
- **Rollout laptop with browser access to the API** — signed in to <https://<your mine API URL>/Admin> as an Administrator account. Confirm Admin → Devices loads before starting.
- **Operator credentials list** — exported from Admin → Employees with employee number, email, and a temporary password for each operator. The operator changes the password on first sign-in.
- **Asset-tag stickers + permanent marker** — every device gets a printed label with its short asset code (e.g. "TAB-OP-04") and the operator's name.
- **The Equipment Checklist mobile app installer** — Android APK (signed by Knox / Intune / sideload-ready), or Windows MSIX for Surface devices.

1.2 What the user needs ready

- Their email address as it appears on Admin → Employees.
- Their temporary password from the rollout sheet.
- Their fingerprint / face already enrolled in the device's OS Settings — they should do this BEFORE handover.

2. Per-device enrolment — the 12-step sequence

Follow these steps in order for each device. The single biggest trap is trying to enable biometric BEFORE the device is approved in Admin → Devices — biometric needs a successful sign-in first, so the device-allowlist step has to land before the biometric step.

#	Action	✓ Done
1	Power on tablet. Confirm OS-level biometric (fingerprint / face) is already enrolled in device Settings.	☐
2	Install the Equipment Checklist app (APK via MDM / sideload, MSIX for Windows).	☐
3	Open the app. On first launch you'll see the Sign-In screen.	☐
4	Type the operator's email + temporary password. Tap Sign In.	☐
5	The yellow 'Device not authorised' callout appears — note the fingerprint string OR scan its QR code.	☐
6	Open Admin → Devices on the web from the rollout laptop. Bulk-Add or Add-One — paste the fingerprint, label the device, assign to the operator (or SHARED for shift-rotation tablets).	☐
7	Back on the tablet, tap Sign In again. The login succeeds and lands on the Dashboard.	☐
8	From the dashboard, return to login (sign out). Tap Sign In once more, but this time tick 'Use biometrics next time' before tapping Sign In.	☐
9	Force-close the app from recents. Reopen it. The OS biometric prompt should fire automatically.	☐
10	Authenticate with fingerprint / face. App opens at Dashboard with operator's machines listed.	☐
11	Set the device label on the back / case with the asset-tag sticker.	☐
12	Hand device to operator with their temporary password printed on the asset-tag receipt.	☐

Note — If you're doing 10+ devices in a row, use the bulk-add CSV form (Section 4) instead of step 6. Capture every fingerprint to a list during steps 1–5 across all devices, then paste the whole list at once. Cuts rollout time roughly in half.

3. Shared vs pinned devices

Each device in Admin → Devices is either pinned to a single operator or marked shared. The choice has security and operational consequences — pick deliberately per role.

Pinned (default)	Device fingerprint is linked to one operator's account. Only that operator can sign in on that device. Lost / stolen device exposure is limited to that one operator's submissions. Best for personal-issue tablets in the cab.
Shared	Device fingerprint is approved, no specific user. Any operator with a valid account can sign in. Best for shift-rotation tablets at handover points (e.g. workshop spare, dispatch counter). Trade-off: anyone with the device + a password can sign in as anyone else with a valid account — biometric is per-user-per-device so the wrong-person risk only matters during the password-typing window.

3.1 Recommended split for a standard mine

Operators (in-cab)	Pinned. The operator's personal Tab Active 5 stays in their cab. The MHSA accountability for pre-shift checks is per-operator-per-shift so pinning matches that legal model.
Supervisors	Pinned. Same reasoning — sign-off is a personal accountability.
Mechanics	Pinned for shift mechanics. Shared for the workshop spare that anyone on call uses.
Spare / pool	Shared. The dispatch-counter Tab S9 FE that any operator grabs during a hot-swap.
Admin / SHE	Pinned to the named admin. Their Surface stays at the office desk.

4. Bulk authorising — the rollout-day shortcut

On day one, you don't have time to enrol each device by tapping through Admin → Add Device twenty times. Use Admin → Devices → Bulk authorise (CSV). The format is:

fingerprint,label,assigned-email

One device per line. Headers and # comment lines are skipped. Blank email means "shared" (toggle in the form changes this default). Example paste:

Operator tablets — pinned

a8e3d7c1...c91f,TAB-OP-01,thuli.dlamini@belfast.co.za

b4f72e90...22a8,TAB-OP-02,sipho.mkhize@belfast.co.za

c1e9f045...8b3d,TAB-OP-03,bongani.nkosi@belfast.co.za

Workshop spare — shared

d6a2c818...440e,TAB-SPARE-01,SHARED

Surface for SHE officer — pinned

fa01b7e3...09bc,SURFACE-SHE-01,she.officer@belfast.co.za

The bulk form will report: how many were added, how many duplicates were skipped (already known fingerprints), how many malformed rows were rejected, and any emails it couldn't find in the user table. Nothing silent — every row is accounted for.

5. Enabling biometric on each device

Biometric is per-device-per-operator. The OS must already have fingerprint/face enrolled (Settings → Biometrics) BEFORE the operator ticks the box in the app.

Step A • First sign-in

On the tablet, after the device is approved, the operator types email + password and ticks "☑ Use biometrics / Windows Hello next time" BEFORE tapping Sign In. The opt-in flag is captured at sign-in, not after.

Step B • Test the prompt

Force-close the app. Reopen it. The OS biometric prompt should fire within a second. Authenticate. The app opens at the Dashboard.

Step C • Verify the manual button

Sign out from the burger menu. Reopen the Login screen. The amber "☑ Sign in with biometric" button should now be visible. Tap it. The prompt fires. This is the manual-trigger path the operator uses every morning.

5.1 What biometric methods are accepted

Fingerprint	Always works on any device with a sensor — Tab Active 5, Tab S9 FE, XCover7, Surface Pro keyboard cover.
Face (Class 3 hardware)	Works on Surface Pro 11 (Windows Hello Face — IR camera). Not available on the recommended Samsung tablets — their selfie cameras are Class 2 and we accept those too now.
Face (Class 2 camera)	Accepted as a fallback on Samsung tablets that enrolled face unlock at OS level. Less secure than fingerprint — possible photo spoof — but acceptable as backup.
Device PIN / pattern	Used automatically as a fallback when biometric fails three times. The operator uses the same number they already type to unlock the phone — they're not stuck.

6. Troubleshooting — when something goes wrong

Symptom	Likely cause	What to do
"Device is not authorised" callout shown after sign-in attempt.	The fingerprint hasn't been added to Admin → Devices yet.	Copy the fingerprint from the yellow callout, paste it into Admin → Devices → Add Device, assign to the right operator or mark SHARED, save, then retry sign-in on the tablet.
Biometric checkbox doesn't appear on the Sign-In page.	The OS hasn't enrolled any biometric yet, OR the device only has Class 2 face unlock and the app's policy excludes it (older builds only).	Open device Settings → Biometrics. Enrol at least one fingerprint. Return to the app and reopen the Sign-In page.
Biometric prompt doesn't fire when the app is reopened.	Either the operator didn't tick "Use biometrics next time" at sign-in, OR the BIO_KEY flag was wiped by a previous sign-out under an old app build.	Sign in with password, tick the box, tap Sign In. Force-close, reopen. The prompt should now fire.
"No saved session on this device" after tapping the explicit biometric button.	Cached session was lost (e.g. uninstall + reinstall, or background auto-sign-out). The auto-heal will clear the inconsistent state silently.	Sign in with password + tick biometric. Confirm the prompt fires on next reopen.
Operator's account has been deactivated message.	Admin has revoked the operator's account from Admin → Employees. This is enforced both online and on the next offline-cache hit.	Confirm with the admin whether reactivation is intended. Admin → Employees → Reactivate clears the block; operator must reconnect once for the mobile to pick it up.
Web biometric button is missing on /Account/Login.	Browser doesn't expose WebAuthn — page is loaded over plain HTTP, or via a bare IP without a trusted cert. Amber callout will say why.	Open the site over HTTPS to a proper hostname (localhost during dev is fine; production needs a real cert). On a self-signed cert in dev, trust it via dotnet dev-certs https --trust.
Mobile app shows offline pill even on Wi-Fi.	App can't reach the API — server URL is wrong (compile-time DevHostIp), or the device is on a different subnet from the API host, or the API is down.	Confirm the operator's tablet can reach the API URL in a browser (https://<api-host>/health if you've added a health endpoint, or simply navigate to the API base URL). If not, check Wi-Fi network selection and the LAN_DEV_RUNBOOK for the pilot environment.
After signing in, operator sees no machines on Dashboard.	Operator hasn't been assigned to any Machines in Admin yet, OR they're signed in as the wrong account.	Open Admin → Employees → click the operator → Assign Machines. Or confirm the operator typed the right email.
Operator tries to start a checklist and sees "You're not currently competent on FEL" (or similar).	Operator's machine-type competency expired or was never granted. MHSA Section 22(a) enforcement is doing its job.	Open Admin → Competencies → click the operator → Add competency for the machine type. Operator's mobile app picks up the change within 5 minutes (periodic /me refresh).
Mobile app crashes on launch after install.	Usually a code-signing mismatch with a previous build, or missing platform permissions.	Uninstall the previous version completely, install the fresh APK / MSIX. On Android, ensure CAMERA + RECORD_AUDIO + USE_BIOMETRIC permissions are granted in app settings.

7. Per-device record sheet

Fill in one row per enrolled device. Keep this sheet — it's the source of truth for the asset-tag stickers and for any future deactivate-and-replace event.

#	Asset tag	Operator name	Email	Pinned / Shared	Date enrolled
1					
2					
3					
4					
5					
6					
7					
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24					

8. After rollout — the first week

Don't disappear after handover. The first week of real use is where the operational rough edges show up.

8.1 Day-1 checks

- Power-cycle each device at end of shift — confirm the next-morning launch fires biometric and lands at Dashboard cleanly.
- Check Admin → Reports for the day. Submissions should be appearing for every operator who completed a pre-shift check.
- Look for any 'NO-GO' entries the SHE officer wasn't notified of — that's a configuration gap to fix (Mine.SheOfficerEmail in Admin → Settings).

8.2 Week-1 review

- Pull the audit trail from Admin → Audit. Filter on user.login + user.logout. Every operator should appear on every shift day they worked.
- Check Admin → Devices → Last Seen. Any device with no activity in 48h is either powered off, lost, or the operator isn't using it — investigate.
- Ask each operator: did the biometric prompt fire reliably? Anyone who says 'I had to type my password every day' has a misconfigured biometric — work through Section 6's troubleshooting with them.

8.3 Decommissioning a device

- Operator leaves / device is lost: Admin → Devices → find the device → Revoke. The next sign-in attempt from that fingerprint is refused.
- Device is being reassigned to a different operator: Revoke the existing entry, add a new one with the new operator pinned. The old fingerprint history stays for audit.
- Device is being temporarily lent to a relief operator: leave the existing pinning alone, give the relief operator a separate temporary credential, OR mark the device shared for the duration.

9. Quick reference card

Print this page separately and pin it next to the rollout workstation.

Web URLs you'll use

/Admin/Devices — the device list
/Admin/Devices/AddDevice — single-device add
/Admin/BulkAddDevices — paste-CSV bulk add
/Admin/Employees — operator accounts
/Admin/Competencies — Section 22(a) certificates
/Admin/Reports — operational dashboard
/Admin/Audit — "who did what, when" trail
/Admin/Settings — runtime configuration

Bulk-add CSV format

fingerprint,label,email-or-SHARED

Blank email = shared (with default toggle on). Lines starting with # ignored.

If biometric won't fire on a tablet

1) Confirm OS-level biometric is enrolled in Settings. 2) Sign in with password, tick the biometric box, tap Sign In. 3) Force-close app from recents, reopen. 4) If still failing, see Section 6 troubleshooting.

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